

Nathaniel Rogers

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EDUCATION

Northwestern University, Evanston, IL

Anticipated Graduation: *March 2027*

Bachelor of Science in Biomedical Engineering | Segal Design Certificate

Overall GPA: 3.93/4.00

Relevant Courses: Interdisciplinary Product Design, Design Thinking and Communication, Design Research, Biomedical Signals & Circuits, Robotics, Quantitative Systems Physiology, Bioelectronics, Materials Science, Biomedical Statistics

SKILLS

Hardware: 3D Printing, Tolerance Analysis, Rapid Prototyping, Circuit Design/Assembly/Debugging, Soldering, FMEA

Software: SolidWorks, OnShape, Python, MATLAB, Office 365, LTSpice, Arduino IDE, HTML, CSS, G*Power, SwiftUI

Communication: Project Management, Public Speaking, Presentation, Technical Writing, Motivation, Time Management

WORK EXPERIENCE

Biomedical Engineering Intern, Garrett Technologies, Inc.

June 2025 – August 2025

- Executed reliability testing on AED charging loops in an environmental chamber (50–60°C), diagnosing PWM instability under thermal stress and informing a firmware update improving performance at rated limit.
- Developed a Python-based tool to decode and visualize ECG, treatment, and AED system logs, generating automated summaries with error translations to reduce analysis time by 15 minutes per log for a team of 9 engineers.
- Collaborated with internal engineering teams and external battery vendors to define acceptance criteria for lithium cell performance under dynamic loads and elevated temperatures, improving supplier reliability.
- Tested 50 PCB trace resistances to ensure quality standard conformance for FDA Class III medical device.

Ambassador, Vera Health

November 2025 – December 2025

- Provided technical and general user support to physicians, students, and industry reps across medical specialties.
- Onboarded 10 new users to a medical literature AI search engine within 3 hours through product demonstration.

LEADERSHIP & PROJECTS

Northwestern University Medical Makers

Evanston, IL

President

May 2025 – May 2026

- Raised \$7000+ in club funding by submitting grant and departmental requests to support student experience.
- Partnering with a multinational medical device company to facilitate 6 project-specific design reviews.
- Secured a new EEG hardware company sponsorship, providing technical support to 60+ student members.

Project Manager: Waterproof Diaphragmatic Pacer

September 2024 – May 2025

- Collaborated with client, Shirley Ryan AbilityLab, to develop a waterproof diaphragmatic pacer system to allow patients to bathe with reduced nurse intervention and health risk.
- Increased attendance 3x while coordinating meetings and design challenges to boost team engagement.
- Presented progress update to 20+ nurses, earning endorsement of solution as enticing to hospitals and home settings.

DSGN 384: Interdisciplinary Product Design

Evanston, IL

Primary Contributor: ENdose – Precision Liquid Medication Dosing System

September 2025 – March 2026

- Applied parametric OnShape CAD modeling and tolerance stack analysis to ensure manufacturability and compatibility across standard ASTM D2911 bottle finishes.
- Rendered 20+ views of the final product using SolidWorks Visualize to support client decision-making.
- Presented final design to a panel of 20+ representatives from industry and academia.